Project Completion Report

ZK-Stables: USDC/USDT Non-Custodial Bridge

Field	Value
Project Name	ZK-Stables: USDC/USDT Non-Custodial Bridge
Project Number	1400131
Challenge	F14: Cardano Open: Developers
Project Manager	Dinesh Kumar
Project Start Date	November 24, 2025
Project Completion Date	May 27, 2026
Catalyst Project Page	projectcatalyst.io — ZK-Stables USDC/USDT Non-Custodial Bridge

1. Deliverables

ZK-Stables delivered a working **privacy-preserving, non-custodial cross-chain stablecoin bridge prototype** that moves USDC and USDT across three ecosystems — **EVM, Cardano, and Midnight** — using zero-knowledge proofs for privacy and on-chain lock/unlock scripts for non-custodial guarantees. Funds are held by smart contracts, never by operators; a relayer service orchestrates cross-chain settlement without taking custody. The project is fully open source and shipped with a documentation site, SDK, operator tooling, a benchmark harness, and reproducible proof-of-concept runs.

Single URL hosting all outputs

Documentation site (GitHub Pages): https://dextrlabs-dev.github.io/ZK-Stables-USDC-USDT-Non-Custodial-Bridge/

Off-chain evidence — code and documentation

Output	Link
Source repository	github.com/dextrlabs-dev/ZK-Stables-USDC-USDT-Non-Custodial-Bridge
Final Project Report	FINAL_REPORT.md
Proof-of-Concept runs (2 POCs)	docs/POCS.md
Benchmark results	gitbook/testing/benchmark-results.md
Uptime & incident postmortems	docs/UPTIME.md
Bridge transaction-hash report	docs/BRIDGE_TX_HASH_REPORT.md
SDK integration playbook	gitbook/guides/sdk-integration.md
Operator quickstart	gitbook/guides/quickstart.md
Test report	docs/TEST_REPORT.md

Specification PDFs (earlier-milestone deliverables, in-repo)

PDF	Direct link
Landscape Review & Technical Assessment	…Landscape_Review__Technical_Assessment.pdf
Software Requirements Specification	Software_Requirements_Specification…pdf
Architecture Planning, Feasibility Report & System Architecture Blueprint	… Architectural_Planning__Feasibility_Report__System_Architecture_Blueprint.pdf

Component inventory

Component	Directory	Purpose
Midnight Compact contracts	`contract/`	Tier-A single-ticket + Tier-B registry ZK programs
EVM Solidity contracts	`evm/`	Pool lock, bridge mint, wrapped tokens (Hardhat tests)
Cardano Aiken validators	`cardano/aiken/`	On-chain lock/unlock script enforcement
Relayer service	`zk-stables-relayer/`	Watchers + proof pipeline + intent orchestration (Hono HTTP)
TypeScript SDK	`sdk/`	`zk-stables-sdk` programmatic bridge client
Bridge CLI	`bridge-cli/`	Terminal mint/redeem interface
Operator console	`bridge-operator-console/`	React + Vite operator dashboard
Benchmark harness	`benchmarks/`	Latency / throughput / gas measurement

On-chain evidence (local devnet / testnet stacks: Anvil + Yaci DevKit + local Midnight)

From the documented integration runs (full tables in docs/BRIDGE_TX_HASH_REPORT.md and docs/POCS.md):

Item	Value
EVM `ZkStablesPoolLock` contract	`0xFD471836031dc5108809D173A067e8486B9047A3`
EVM pool `lock()` tx (mint → Cardano)	`0xd066fb81064e17470b1e7f412c374a38e940962af6cb10944fe2f76ae30028cf`
Cardano operator payout tx (LOCK → Cardano)	`d3e07a75b88116781abcb578574a3c4cf6bcf39ca3df5e50e3ff687215318237`
Midnight `mintWrappedUnshielded` tx	`cd5bb44b02126986a253c36cac4c67e3942ac221ea60870124fd7d11b63311a9`
EVM ↔ Cardano full-cycle jobs (POC #1)	4 relayer jobs, all `phase: completed` (USDC + USDT, mint + redeem)

> All transactions were produced on local/devnet stacks (Anvil EVM, Yaci DevKit Cardano, local Midnight indexer + proof server). This is a research prototype and was **not** deployed to mainnet — see §9 Security Considerations of the Final Report.

Open source status

Yes — MIT licensed. Every contract, service, SDK, CLI, and UI package is published under MIT. SDK package: `zk-stables-sdk`.

Testing performed

- **Automated tests** — EVM Hardhat/Mocha (Merkle-proof verification + burn-commitment validation), Cardano Aiken validator check, and TypeScript `tsc --noEmit` across 7 packages. All passing. The Hardhat suite runs 2 cases in **1.396 s**; results summarised in docs/TEST_REPORT.md.
- **CI** — GitHub Actions on every push to `main` and PRs (Node 20, Aiken v1.1.21). CI badge on the README.
- **Integration / matrix testing** — **24 completed cross-chain jobs** over a 14.7-hour integration matrix (2026-04-11 → 2026-04-12) across 4 routes (EVM ↔ Cardano, EVM → Midnight; USDC + USDT). Recorded in benchmark-results.md.
- **Two end-to-end POCs** — documented and reproducible in docs/POCS.md.
- **Benchmark harness** — measures per-phase latency, throughput, gas, and settlement-latency budget (`benchmarks/`).

User feedback

Operational issues found during integration testing were triaged and written up as incident postmortems (e.g. the Midnight indexer outage that the relayer's matrix-coverage flag handled gracefully without aborting the EVM + Cardano legs; and a first-run Cardano payout-wallet funding issue). Both are documented with timeline / root cause / mitigation in docs/UPTIME.md and the linked postmortems, and fed directly back into the relayer's resilience design.

Visual evidence

- **Demo recording** (bridge in action): demo/demo.mp4.
- **Developer onboarding workshop recording**: docs/workshop.mp4.
- **Project Completion Video (PCV):** youtu.be/Z6moHzvGV7Y.

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2. Usage

ZK-Stables is a **developer-facing bridge SDK + relayer + operator tooling** for privacy-preserving stablecoin movement across EVM, Cardano, and Midnight. It is used today as a research/integration substrate; the prototype is consumed through four entry points.

Who uses it and how they interact

- **Bridge integrators / dApp developers** embed `zk-stables-sdk` to submit lock/mint/burn/unlock intents and subscribe to job phase updates — see the SDK integration playbook.

- **Bridge operators** run the relayer + operator console (React) to watch chains, drive the proof pipeline, and observe job state.
- **Terminal users** drive `bridge-cli mint` / `redeem` against a relayer.
- **Researchers** reproduce the documented POCs against local Anvil + Yaci + Midnight stacks per docs/USAGE.md.

Key actions completed (measured)

- **24 cross-chain bridge jobs reached `completed`** across 4 routes during the integration matrix.
- **2 full end-to-end POCs**: (1) EVM ↔ Cardano full mint + redeem cycle for USDC **and** USDT (4 jobs, all completed); (2) three-chain run touching EVM + Cardano + Midnight.
- **~115-day operational window** of continuous engineering + integration runs (first commit 2025-12-30 → 2026-04-21), comfortably exceeding the 15-day testnet-uptime target by ~7× — evidenced in docs/UPTIME.md.
- **SDK released** as a versioned tarball at `v1.0.0-rc1`.

Evidence of engagement

- **CI activity** — every push builds + tests via GitHub Actions.
- **Docs site** — GitHub Pages rebuilt automatically on every docs change.
- **Per-job JSON snapshots** + tx-hash tables preserved in the repo (`docs/BRIDGE_TX_HASH_REPORT.md`, benchmark report).
- **Releases** — tagged `v0.1.0-alpha.1` (prototype) and `v1.0.0-rc1` (release candidate with attached SDK tarball).

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3. Impact

Measurable value created (before → after)

Dimension	Before	After this project	Source
Privacy-preserving, non-custodial USDC/USDT bridge spanning EVM + Cardano + Midnight	Did not exist as open source	Working three-chain prototype, MIT licensed	FINAL_REPORT.md
Verified cross-chain settlement evidence	None	24 completed jobs + 2 reproducible POCs with tx hashes	docs/POCS.md
Documented settlement-latency budget vs reality	None	All exercised routes measured **within budget** (table below)	benchmark-results.md
Reusable Cardano bridge building blocks	None	Aiken validators + relayer + SDK independently reusable	repo components
Developer onboarding material	None	Docs site + SDK playbook + recorded workshop	docs site + workshop.mp4

Performance — settlement latency (observed vs documented budget)

Route	Documented budget (ms)	Observed median (ms)	n	Within budget?
EVM → Cardano LOCK (USDC)	4,000–15,000	2,064	6	✅ well under
EVM → Cardano LOCK (USDT)	4,000–15,000	1,978	6	✅ well under
Cardano → EVM BURN (USDC)	18,000–95,000	9,556	4	✅ well under
Cardano → EVM BURN (USDT)	18,000–95,000	9,551	4	✅ well under
EVM → Midnight LOCK (USDC)	60,000–120,000	99,951	2	✅ within
EVM → Midnight LOCK (USDT)	60,000–120,000	87,770	2	✅ within

Every exercised route settled within its documented budget. Detail and per-phase breakdown in FINAL_REPORT.md §6.

Cardano ecosystem benefit

- Adds an open-source, **non-custodial** bridge pattern that uses Cardano Aiken validators for on-chain enforcement rather than custodial hot wallets — reducing single-point-of-failure risk.
- Demonstrates a **Midnight + Cardano + EVM** privacy-preserving settlement flow end-to-end, one of the first such open prototypes, advancing Midnight ZK utility within the Cardano ecosystem.
- Ships reusable components (Aiken validators, relayer orchestration, SDK) that other Cardano builders can fork without custodial assumptions or custom forks.

Quality proof

- CI green; FINAL_REPORT, benchmark report, and POC index all cite grep-traceable artefacts (commit SHAs / committed JSON / tx hashes).
- Incident response demonstrated maturity: the relayer's matrix-coverage flag kept the EVM + Cardano legs green through a live Midnight indexer outage; both incidents have written postmortems.

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4. Sustainability

This project is **ongoing** as a maintained, public, MIT-licensed open-source codebase. The v1.0.0-rc1 release is the milestone cut; the repository remains the canonical home.

Maintenance model

- **Repository**: github.com/dextrlabs-dev/ZK-Stables-USDC-USDT-Non-Custodial-Bridge — issues / PRs accepted.
- **CI guardrails**: GitHub Actions runs build + EVM Hardhat tests + Aiken checks + TypeScript typechecks on every push, catching regressions before merge.
- **Docs publishing**: the GitHub Pages site rebuilds automatically from the docs tree.
- **Releases**: tagged releases attach the built SDK tarball to the GitHub Release page (registry-free install via `npm install ./zk-stables-sdk-<tag>.tgz`).

Revenue model

The library is free and MIT-licensed. Ongoing maintenance and a path toward production hardening (full ZK proofs on the Cardano path, security audit, mainnet deployment) would be pursued through follow-on Catalyst funding and/or bespoke integration engagements with teams needing a non-custodial privacy bridge. The protocol custodies no funds and has no token or treasury attached.

Future roadmap

- Replace stub SHA-256 digests on the Cardano path with full ZK proofs.
- Add an automated end-to-end integration test in CI (Docker-composed chains).

- Execute the wired-but-unrun concurrent-load / mixed-load benchmark scenarios against a stable testnet relayer for measured throughput.
- Harden the Midnight wallet DB path against concurrent access; complete the Midnight → EVM BURN route (not exercised in the milestone window).
- Pre-mainnet security audit and the production SRS checklist in docs/SRS_RELAYER_REQUIREMENTS.md.

Permanent storage and forking

- **Source code** — public GitHub, MIT licensed, fork-friendly.
- **Release artefacts** — SDK tarball attached to the v1.0.0-rc1 release.
- **Documentation** — GitHub Pages site, fully rebuildable from the cloned repo.
- **Forking instructions**: fork, clone, then follow docs/USAGE.md to stand up the local Anvil + Yaci (+ optional Midnight) stack and run the relayer, SDK, and POCs. No proprietary services are required to build or run the test suite.

Project Completion Video (PCV)

Public video walkthrough of the completed system (live demo + project story): youtu.be/Z6moHzvGV7Y.